04 DEPT - Vehicle Dynamics

SRP-038 – Brakes

Development Process

Sunswift Vision Statement:

Sunswift Racing 8 is a high-performance prototype vehicle that aims to showcase as many cutting-edge technologies as possible. It is versatile in providing a refined driving experience both on the road and around a racetrack. Overall, the car has a high excitement factor that entices people to drive it, defeating the misconception that solar electric vehicles cannot be interesting and engaging.

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II. Document Revisions

Document Version	Date	Description	Approved by
0.1	30/10/2022	Conceptual Design	Declan Watson
0.2		Preliminary Design	

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1 - Conceptual Design

1.1 - Projects

Project Scope

SRP – 038 is responsible for the development of the Brakes subsystem acting within Vehicle Dynamics. This will consist of from (not including) the pedal through the hydraulic lines, master cylinder and brake booster, through to the callipers and brake discs. Also, Brakes will be responsible for the design of the park brake. The aim of the brakes department is to research viable braking systems and options in order to determine packaging requirements, fundamental for other vehicle systems and importantly the design of suspension uprights.

Terms, Definitions, and Abbreviations

Table 1: Terms, Definitions, and Abbreviations

Abbreviation/Terms	**Definition**
TPM	Technical performance measures
SR8	Sunswift 8
OTS	Off-the-Shelf

Responsible and supporting departments

Vehicle Dynamics (Responsible)
Interior (Supporting)

Project interfaces

SRP – 038 will interface with the following departments:

Table 2: Terms, Definitions, and Abbreviations

Relevant Departments	**Description**
SRP – 010 Test Mule	Test mule has accurate dimensions (pickup points and barking system dimensions) for effective testing and optimization.
SRP – 022 Human Machine Interface	Braking controls that can be adjusted via interfaces on the steering wheel such as brake bias.
SRP – 026 Pedals	Providing interior with specific details such as type of pedal as well as percentages of pedal travel
SRP – 035 Performance optimisation	Braking performance can be adjusted to suit conditions, optimize braking distance, and minimize braking temperatures.
SRP – 040 Steering	Ensuring that braking components (eg lines) do not interfere with steering components.
SRP – 041 Suspension	Pick up points and locations that will be necessary for the construction of uprights.

SRP – 042 Wheels	Ensure that brake disks and calipers fit within the inside barrel of the wheel.
SRP8 - 015 Cruise Control	If it is adaptive, brakes must be used when distance shortens between the car and the object in front of it
SRP8 - 098 Automated Parking	Brakes must be used
SRP8-059 Sensors	Sensors will be used to monitor pressure and collect data
SRP8-014 Regen Braking	Regen braking system must be implemented into the system

Assigned team members

- Martin Nguyen (Project Lead, 24T1 – Present)
- Declan Watson

1.2 - Design Factors

Lifecycle factors

The brake system in theory should have a lifespan which mimics that of the vehicle. Disposable components of the braking system include hydraulic lines, piston seals, brake pads and brake discs.

It should be considered throughout preliminary design as the braking system should last over the average lifespan of the typical passenger vehicle.

Typically for passenger vehicles this lifespan is 90,000 km's for brake pads and 150,000 km's for brake discs.

Risk Analysis

Major concerns would surround the braking force and impact of heat. Brakes when heated (more specifically overheated) lack braking performance.

Typical braking performance should be compared among varied materials to optimize braking performance and vehicle safety.

Table 3: Risk Analysis Table

Risk	Response Category	Mitigation Strategy
	Braking Hardware	
Brake Fade	Mitigate, Avoid	Focus on a solution which dissipates heat. Reduce the number of heavy braking intervals while on track.
Brake Pad (and Disc) wear	Accept	Regular servicing/inspections which is included as a vehicle inspection check. See lifecycle for lifespan of these components.
Brake squeaks	Mitigate	Every time the brakes have come into contact with someone, they are to be cleaned with Brake Cleaner.
	Braking System	
Brake leak	Mitigation	Use of braided lines as well as regular inspections of the braking hardware to mitigate splits in brake lines.

		The use of Teflon tape at all fittings.
Brake Booster fails	Servicing, Mitigation	If the Brake Booster is to fail the system will be bypassed and intermechanical pressure will remain between driver and braking hardware
Regen		

Commented [DW2]: talk to pwrtn about regen specifications

Innovation Analysis

Material analysis, lifecycle analysis and comparisons will need to be conducted to determine feasible options but also options which are worthwhile.

Commented [JT3]: Might need to do a lifecycle analysis

This would include the possibility of manufacturing in house of components to justify some decisions.

It is important to keep in mind that by trying to find innovative solutions it would be best for us to manufacture them rather than buying them off-the-shelf. By doing this factor of safety will be impacted which must be tested to ensure the upkeep of both legal standards but also road-user safety.

Manufacture and Procurement

This would include the exploration of production of braking components such as brake discs. This would be with the help of McConaughy. Otherwise, off-the-shelf options must be explored.

With the braking system being such a fundamental factor of safety of a passenger vehicle it would require a high level of attention and strict procedures if it were to be manufactured.

Table 4: Manufacture and Procurement

Item	Method
Mounting Bell	In house CAM or send to Sorensen or can be purchased OTS.
Discs	Purchased OTS

Design and Production plan

Key milestones include:

- Preliminary Design
- Detailed Design
- Construction

Cost Analysis:

The main components that make up the bulk of the cost for the brake system include the callipers, brake discs, master cylinders, IBSe9's and electronically activated callipers due to their complex design. The cost of the parts can be reduced using partnerships with AP Racing, Comp Friction and LSP. The table below provides an estimation of the costs of each brake system component based on information provided by distributers online.

Component	Component Code	Quantity	Link	Cost per unit in AUD (Based on approximate prices available)	Total
Brake callipers	AP Racing CP9200-2S0BG	2	https://www.demon-tweeks.com/au/ap-	$963.07	$1926.14

			racing-cp9200-4-piston-two-piece-forged-radial-mount-brake-caliper-7000565/		
Brake callipers	AP Racing CP9200-3S0BG	2	https://www.demon-tweeks.com/au/ap-racing-cp9200-4-piston-two-piece-forged-radial-mount-brake-caliper-7000565/	$963.07	$1926.14
Brake discs	AP Racing CP3047-252/253	4	https://www.demon-tweeks.com/au/ap-racing-ventilated-brake-discs-300-355mm-240005/	$754.46	$3017.84
Brake pads	AP Racing CP3215D50	4	https://intimabrakes.com.au/product/intima-sr-md8472-sr/	$199	$796
Master cylinders	CP7854	2	https://www.murraymotorsport.com/ap-racing-cp7854-master-cylinder-trunnion-mount-aprcp7854-c	$615.33	$1230.66
Reservoir	CP2293-141/-143	2	https://www.rpxpress.com/products/reservoir-single-90-outlet-ap-racing-cp2293-143	$97.43	$194.86
Brake Lines	-	-	https://aeroflowperformance.com/brake/hose-fittings/hardline	$7.51/m	$57.05
Brake Hose	-	6	https://www.vsport.com.au/?rf=kw&kw=brake+lines	$240	$1440

Brake line pressure sensors	-	2	https://microtechefi.com/product/pressure-sensor-0-1500/	$125	$250
Residual pressure valve	-	2	https://www.proflow.com.au/brake-residual-valve-2lbs-blue-1-8-npt	$21.22	$42.44
Electronic Park Brake (right)	Wilwood 120-15702	1	https://wilwood.com/calipers/CaliperProd?itemno=120-15702-BK	$563.21	$563.21
Electronic Park Brake (left)	Wilwood 120-15703	1	https://wilwood.com/calipers/CaliperProd?itemno=120-15702-BK	$563.21	$563.21
Balance Bar	CP5520-25	1	https://apracing.com/race-car/pedal-boxes/balance-bars/trunnion-types/conventional-mount-high-efficiency-type-cp5520-25	Already owned	-
Brake Booster	IBSe9	2	https://www.lsp-ias.com/our-products/ibse-motorsport-systems	Already owned	-
Total					$12007.55

Educational Outcome:

If manufacture were to be viable and safety is a factor that can be upheld the process of manufacture would provide an exceptional learning opportunity for Vehicle Dynamics and more broadly the Mechanical team.

This learning experience would require communications with McConaughy and potentially Sorensen to ensure that safety items are met.

The students will be able to:

- Analyze braking performance and assemble braking systems
- Research and source off-the-shelf braking components
- Collaborate efficiently with other departments such as Chassis & Bodywork, Interior

Upskilling Development:

Upskilling development is all dependent on whether manufacture is viable. If so inductions in Mill and Lathe would be necessary.

- Sheet Metal – Induction needed before mill and lathe inductions
- OptimumG Series
    - OptimumTire
    - OptimumK

1.3 - Conceptual Requirements

Functional requirements

Table 5: Functional requirements table (Normal conditions)

Reference	Title	Description	TPM
REQ1.01	Vehicle Stops	Braking capacity meets standard from calculations.	TPM101 – The braking system stops the vehicle from moving.
REQ1.02	Vehicle Packaging	Brake disc and caliper fits within the barrel of the wheel	TPM102 – Braking package fits within the wheels.
REQ1.03	Vehicle Packaging	Brake lines and master cylinder do not impede other interfaces within the vehicle.	TPM103 – Brake lines and master cylinder have specified locations.
REQ1.04	Vibrations	Withstands all road vibrations.	TPM104 – Brakes handle all road vibrations caused by tarred road travel (Track and public road)
REQ1.05	Braking	Limited brake fade (consistency)	TPM105 – Brake fade is minimal and does not require interruptions during track runs.
REQ1.06	Braking	The brakes should not differ (at least to the feel of the driver) in the wet/dry or cold/hot.	TPM106 – Braking performance is equivalent and consistent despite conditions.
REQ1.07	Service	Braking package can be serviced easily without having to remove multiple parts of the vehicle.	TPM107 – Braking package has easy access for service.
REQ1.08	Enironmental	The system is clean and doesn't leave behind copious amounts of brake dust.	TPM108 – System has an improvement on ecological cleanliness compared to current passenger vehicles
REQ1.09	Packaging	The entire braking package is lightweight and compact	TPM109 – The entire braking package is compact and lightweight.
REQ1.10	Fail Safe	Two separately plumbed systems can stop the car	TPM110 – There are two systems to stop the car in case of failure.

Performance requirements

Table 4: Performance requirements (Teamwide specifications)

Table 6: Performance requirements (Teamwide specifications)

Reference	Title	Description	TPM
REQ2.01	Unsprung mass	Brakes weight should be minimal in order to reduce unsprung mass and enable faster reaction times for SR8 suspension.	TPM201 – Brakes weight should be optimized to be as small as possible.
REQ2.02	Rolling resistance	Rolling resistance of the vehicle should be nominal. Allowing greater efficiency and performance. Brake disc weight and pad retention are considerations for this.	TPM202 – Rolling resistance should be optimized to be nominal.
REQ2.03	Safety	The braking system is responsible for the safety of the occupants and must safely stop the vehicle	TPM203 – Brakes safely bring the vehicle to a stop.
REQ2.04	Braking Capacity	Braking capacity should be made appropriate to meet all functional requirements and as many performance requirements as possible.	TPM204 – Braking capacity should be made to meet as many performance requirements as possible while still meeting all compulsory requirements.
REQ2.05	Track performance	Brakes should optimize performance around the track. In accordance with the SR8 vision.	TPM205 – Brakes are optimized for track performance in the pursuit of the best lap times.
REQ2.06	Road efficiency	Brakes are designed for road driving and optimized for this use case.	TPM206 – Brakes are optimized for road driving in terms of safety, efficiency and durability.
REQ2.07	Innovation	The braking system must be innovative and create excitement around SR8.	TPM207 – Braking system has innovative ideation and distinguishes us from other solar car projects.
REQ2.08	Sustainability	Brakes must be as sustainable as possible in terms of material use and manufacturing. End of life cycles are considered for brake discs and pads.	TPM208 – Brakes are designed to be sustainable.

Interface requirements

Table 5: Interface requirements (Project/System interfaces)

Table 7: Interface requirements (Project/System interfaces)

Reference	Title	Description	TPM
REQ3.01	Packaging	Interface between projects. Close collaboration with Packaging to enable other departments to interface with brakes and to determine the space in which brakes fit and operate.	TPM301 - Packaging requirements for brakes are met to ensure no impedance to braking performance and drivability of other departments.
REQ3.02	Test Mule	Test mule has accurate dimensions (pickup points and braking system dimensions) for effective testing and optimization.	TPM302 - Test mule has accurate dimensions (pickup points and braking system dimensions) for effective testing and optimization.
REQ3.03	Human Machine Interface	Braking controls that can be adjusted via interfaces on the steering wheel such as brake bias.	TPM303 – Braking controls that can be adjusted via interfaces on the steering wheel such as brake bias.
REQ3.04	Pedals	Providing interior with specific details such as type of pedal as well as percentages of pedal travel.	TPM304 - Providing interior with specific details such as type of pedal as well as percentages of pedal travel.
REQ3.05	Steering	Ensuring that braking components (e.g. lines) do not interfere with steering components.	TPM305 - Ensuring that braking components do not interfere with steering components.
REQ3.06	Suspension	Pickup points and locations that will be necessary for the construction of the uprights.	TPM306 – Discussions about pickup points and locations that will be necessary for the construction of the uprights.
REQ3.07	Wheels	Ensuring that the brake discs and calipers fit within the inside barrel of the wheel.	TPM307 - Ensuring that the brake discs and calipers fit within the inside barrel of the wheel.

External requirements

Table 6: External requirements

Table 8: External requirements

Reference	Title	Description	TPM
REQ4.01	Regen and Mechanical Braking (ADR 2.15 & 2.17)	Brakes function both under regen and mechanical braking.	TPM401 – Brakes function both under regen and mechanical braking.

REQ4.02	Vibrations (ADR 5.1.1.1.)	Brakes continue to function without change under vibrations.	TPM402 – Brakes continue to function without change under vibrations.
REQ4.03	Fail Safe (ADR 5.1.2.2.)	There is a secondary braking system in play in the vehicle.	TPM403 – There is a secondary braking system in play in the vehicle.
REQ4.04	Park Brake (ADR 5.1.2.3.)	The park braking system functions at specific angles of inclination and declination.	TPM404 - Park braking system functions at specific angles of inclination and declination.

Notes: vehicle category code defined by ADR: “MA” for passenger car. ADR01/00 – 01 designates number, 00 designates revision.[ADR's (M1): ADR31.pdf]

2 - Preliminary Design

2.1 - Key Components

Key Components definition

Facilitate the movement of SR8 while supporting the load presented by all subsystems. Primarily applying the torque or braking forces applied to SR8 onto its environment as required. Maintaining energy efficiency, stability and limiting rolling resistance.

Project:	Front Brakes
Key Components:	Brake calipers (inc. Dog bone) Brake disc (inc. Mounting bell) Brake pads Master cylinder Reservoir IBSe Hard brake lines Braided brake lines Brake fluid Fittings (banjos, elbows, t-pieces)

NOTE: Balance Bar is included yet it is what interfaces between front brakes and rear brakes.

Project:	Rear Brakes
Key Components:	Brake calipers (inc. Dog bone) Brake disc (inc. Mounting bell) Brake pads Master cylinder Reservoir IBSe Hard brake lines Braided brake lines Brake fluid Fittings (banjos, elbows, t-pieces)

Project:	Park Brake
Key Components:	Brake Caliper (separate caliper potentially which is then added to the rears) Actuation (electric motor)

2.2 - Design Factors

Lifecycle

Components are considered to be in one of three categories:

1. Off-the-shelf components: Will have a lifecycle revolving around product selection.
    - Product Selection
    - Fitment & Assembly
    - Testing & Maintenance
    - Afterlife
2. House-made components: Will have a lifecycle revolving around design and production.
    - Design & Analysis
    - Machining
    - Fitment & Assembly
    - Testing & Maintenance
    - Afterlife
3. Repurposed components: These are components already used and/or purchased which will be used for SR8
    - Collation of Data
    - Review of previous life
    - Design Integration
    - Fitment & Assembly
    - Testing & Maintenance
    - Afterlife

Key component	Lifecycle
Brake Calipers	Cat. 1: These will be purchased from the AP Racing catalog range and follow the recommended service life maintenance schedule.
Dog Bone	Cat. 2: These will be made to preserve the calipers and the uprights. This will be made from an alloy and anodized for increased life.
Brake disc	Cat. 1: These will be purchased from the AP Racing catalog range and follow the recommended service life maintenance schedule.
Mounting Bell	Cat. 2: This will be made to promote the life of the discs and calipers which are often impacted by heat. This will be made from an alloy and anodized for increased life.
Brake pads	Cat. 1: These will be purchased from the AP Racing catalog range and follow the recommended service life maintenance schedule.
Master cylinder	Cat. 1: These will be purchased from the AP Racing catalog range and follow the recommended service life maintenance schedule.
Balance bar	Cat. 3: An unused part which was additionally purchased. This part is from the AP Racing range and will interface with all AP Racing components.
IBSe	Cat. 3: Previously used for the Remote Driving project the IBSe's will be the brake-by-wire interface responsible for also boosting the brake pressure.
Reservoir	Cat. 1: These will be purchased from the AP Racing catalog range and follow the recommended service life maintenance schedule.

Hard brake line	Cat. 1: These will be purchased from the AP Racing catalog range and follow the recommended service life maintenance schedule.
Braided brake line	Cat. 1: These will be purchased from the AP Racing catalog range and follow the recommended service life maintenance schedule.
Brake fluid	Cat. 1: These will be purchased from the AP Racing catalog range and follow the recommended service life maintenance schedule.
Fittings	Cat. 1: These will be purchased from the AP Racing catalog range and follow the recommended service life maintenance schedule.

Risk

- Risk
  - **What's Required?**
    - All components must have individual associated risk and mitigation strategy
    - Comparison against the conceptual phase predictions
    - The discussion and reflection about encountered risks and implemented solutions up to this stage are encouraged for the presentation
  - **Additional Notes**
    - Highlighted risk factor includes of
      - Possible delays
      - Possible failures
      - Unknowns
    - Risk analysis can focus on risks throughout the lifecycle of the component, from design, manufacture, and final application

Innovation

The brakes project has experienced low to medium innovation. A number of the key components have been analysed, compared to different options and justified. A few key components will be manufactured impacting the factor of safety which will need to be tested to ensure legal standards and road-user safety is kept.

- Brake discs – medium innovation
  - Bolted disc was chosen off the shelf
  - Manufacture of the mounting bell – medium innovation
  - Reason: Allows for student involvement
- Handbrake – high innovation
  - Hand brake chosen overws2a electronic brake
  - Manufacturing in house
  - Reason: Allows for student involvement
- Hard and braided brake lines chosen – low innovation
  - Will be sourced off the shelf
  - Reason: manufacturing not possible by students
- Pull type master cylinder – low innovation
  - Either CP6461 or CP6465
  - Sourced off the shelf from AP Racing
  - Reason: technical limitations (high failure risk/safety issues)
- Front and rear callipers – low innovation

Commented [JT4]: High Innovation or high education

    - Chosen off the shelf from AP Racing
    - https://apracing.com/race-car/brake-calipers/pro-5000-r-range/4-piston-cp9440-2-3s4l-front-4712cm2-piston-area?switch=0
    - https://apracing.com/race-car/brake-calipers/pro-5000-r-range/4-piston-cp9441-2-3s4l-rear-3619cm2-piston-area?switch=0
    - Reason: manufacturing not possible by students
- Brake pads – low innovation
    - Sourced off the shelf from AP Racing
    - https://apracing.com/race-car/brake-pads/4-piston-caliper-pad-profiles/cp3215d50
    - https://apracing.com/race-car/brake-pads/4-piston-caliper-pad-profiles/cp3215d50
    - Reason: manufacturing not possible by students

- Innovation
    - **What's Required?**
        - Analysis of each key component based on Section 3.2.3
        - Revise the project's innovation analysis and identify any changes
        - Prepare justification for each outcome
        - The ideal outcome would be same rank or even higher when compared to Conceptual Design
        - Example:
            - Door (Project) -> Medium Innovation
            - Door Hinge (Key Component) -> Low Innovation
            - Unsatisfactory according to the analysis from the previous phase
            - Reason -> Technical limitations (high failure risk/safety issues)
    - **Additional Notes**
        - To effectively perform the analysis, the actions below are a good start:
            - Identify new designs
            - Identify new risks
            - Determine the result of the risk and innovation matrix
        - Some technical sketches or visual aids are compelling justifications for risk/innovation justification

Manufacture and Procurement

- Manufacture and Procurement
    - **What's Required**
        - The components are expected to be individually designed and manufacture, therefore, each must have a customized plan
        - After customizing, a revision of the conceptual design plan is necessary to prevent significant variations and replanning
        - Justification of significant changes is important, identify any necessary considerations regarding companies or suppliers

- Example:
  - Supplier list for project A -> company X / Y
  - Supplier list for component A.1 -> company X / Z
  - Company X is the only partnered company with well-established relations and gets listed on the previous plan
  - Company Z is the new identified partnered company that is necessary for the success of component R&D
    - The team has no connections with the company
    - Expecting significant collaboration and technical support throughout the manufacturing and machining process

- **Additional Notes:**
  - Possible Considerations
    - List of Suppliers
    - Industry relations
    - Plans for component acquisition

Design and Production Plan

- Design and Production Plan
  - **What's Required**
    - Revise previous planning and understand the objectives
    - Construct an individual plan for components according to the project timeline
    - At this stage, the plan will focus on detailed design and manufacture
    - Example
      - Door (Project) -> 8 months
      - Key Components
        - External body -> 2 months
        - Interior -> 2 months
        - Electricity system -> 2 months
        - Software system -> 2 months
  - **Additional Notes**
    - Points that should provide adequate information while
      - Match timelines decided in the conceptual phase
      - Expected time for the following phase
      - Project milestones and objectives
      - Identify high lead time components or processes
    - Exemptions
      If there are any high lead-time components, early ordering may be necessary before moving to the next phase to avoid delays. However, strong reasoning and evidence must prepare to support such a decision. The evidence could include:

      - Market analysis
      - Information from supplier

Cost Analysis

- Cost Analysis
  - **What's Required**
    - Each main component must have quantitative estimation with high accuracy
    - High accuracy -> supported by relevant documentation (estimate quote from suppliers)
  - **Additional Notes**
    - Expected outcome
      - Estimation of cost for components
      - Identification of unknown costs
      - Reasoning for unknowns
      - Categorization of costs
    - Categories for costs
      The cost could be identified as "actual cost" or "cost to Sunswift". The table below shows the method to differentiate both.

Type	Definition
Actual Cost	Standard cost paid for the component (High accuracy is encouraged)
Cost to Sunswift	Forecast cost for Sunswift

Technology that is wholly provided from companies will add to the "actual cost" tally.

Educational Outcome

- Educational Outcome
  - **What Required?**
    - Discuss how much you have learned from designing and component acquiring
    - Example
      - Learnt negotiation skills from communications with suppliers
      - Metal 3D printing -> machine operations and design
      - Understand types and implementation of sustainable materials
  - **Additional Notes**
    - The discussion is not limited to technical skills and can include
      - Verification of design factors
      - Industry relations
    - This section mainly assesses the design engineers' understanding and thoughts about the consequences of every planned and current task from an undergraduate perspective

Upskilling Development

- Upskilling Development
  - **What Required?**
    - Re-identify the difficulties and challenges that the component may encounter
    - Shortlist and order the actions that may take place before further progression
    - Reflect on the advantages of the expected upskilling

2.3 - Preliminary Requirements

Functional Requirements

3.1.1. Functional Requirements

Define the operational functions that the vehicle should achieve in various environments to fulfil the user needs in all scenarios.

1. Normal Operation:
   Expected operation with all components functioning as intended.

2. Abnormal Operation:
   a. User Error:
   Where error from the user may cause illogical operation or inputs. (e.g., pressing pedals at the same time)

   b. Emergency Scenario:
   Where error from the environment may cause damage or abnormal operation. Encompasses crash scenarios and component/subsystem failure.

3. Testing / Maintenance / Fault Diagnostic Operation:
   Functionality which is specifically designed for testing, maintenance, or fault diagnosis operations, independent of the vehicle's explicit operation.

Emergency Scenarios

User Error

Testing/Maintenance/Fault diagnostic

Performance Requirements

3.1.2. Performance Requirements

Define the extent to which a function must be executed.

1. Teamwide Specifications
   High level benchmarks for the overall of performance of Sunswift 8 set out with the team vision.

2. Technical Specifications
   A broad range of technical requirements at the vehicle, subsystem, and component level, for example, size, weight, data sampling rates, latency.

3. Safety Specifications
   Requirements independent of the systems intended functionality which improve the safety of the vehicle.

Teamwide specifications

Technical specifications

Safety specifications

Reliability Requirements

3.1.4. Reliability Requirements

Any general considerations in the reliability and consistency of components which will be worn down over time. E.g., electrical connector ratings, mechanical component lifespan etc.

Interface Requirements

3.1.3. Interface Requirements

Define how independent projects and subsystems are connected.

1. Project/System Interfaces

   Explicitly define the boundary between two projects or two subsystems; define where the boundary of each system is and how they are connected. Can be specific, for example, connector types.

2. Mounting Points/Areas/Routing

   Defining the volume/area left for other project components, must be agreed upon between both project's members.

Project/System interfaces

Mounting points/Area/Routing

External Requirements

3.1.5. External Requirements

Requirements which originate from outside the team, because of team goals such as road legality or competitive compliance.

1. Australian Design Rules (ADRs)

   Under the vehicle classification: MA (Passenger Car), there are a variety of standards which the car must adhere to.

2. ISOs

   Non enforceable standards under Australian law and as act only as guidelines for the Sunswift 8 build process. They are being used to improve the safety of the electric drivetrain and fill in any gaps with ADR's. The standards that are being used fall under the heading ISO 43.120, for electric road vehicles, and are:

   i. ISO 6469 – Electric Road Vehicles Safety Specifications
   ii. ISO 17409 – Conductive Power Transfer Safety Requirements
   iii. ISO 18300 – Test Specifications for Lithium Batteries
   iv. ISO 21498 – Electrical specifications and tests for voltage class B systems and components
   v. ISO 21782 - Test specification for electric propulsion components.
   vi. IEC 62752 – In cable control and protection device for mode 2 charging of electric road vehicles

3. UNECEs

   The UNECE standards are a superset of ADR's and precede most ADR changes. They are not enforceable under Australian Law. They can be used in a similar manner to ISOs. The standards that are being used fall under UNECE WP.29, and are:

   i. WP29 10 – Approval of vehicles with regard to electromagnetic compatibility

ii. WP29 94 - Approval of vehicles with regard to protection of occupants in the event of a frontal collision
iii. WP29 95 – Approval of vehicles with regard to protection of occupants in the event of a lateral collision
iv. WP29 153 - Approval of vehicles with regard to fuel system integrity and safety of electric power train in the event of a rear-end collision

Design comparison

References

Quiroga, T. 2010, Effects of Upsized Wheels and Tires Tested, *Car and Driver*, <*https://www.caranddriver.com/features/a15130598/upsized-wheels-tires/*>